

blockchain. Nodes running the old software will reject blocks created under the new rules, and vice versa. This results in two separate blockchains that share a common history up to the fork point but continue independently afterward.

Hard forks can occur for various reasons:

- **Protocol upgrades:** When developers implement significant improvements that require breaking changes to the blockchain's rules.
- **Community disputes:** When disagreements about the future direction of a blockchain can't be resolved, resulting in a split into two separate projects.
- **Security incidents:** In response to major security breaches or exploits, communities may choose to fork into a new chain that addresses vulnerabilities.

The technical process of executing a fork involves coordinating changes across the network's nodes at a specific block height. This requires careful planning and communication among developers, miners or validators, and users to ensure a smooth transition

Forking as a catalyst for innovation

Forking has proven to be a powerful mechanism for driving innovation in blockchain technology. By allowing for experimentation and iteration without requiring unanimous consensus, forks enable technological progress that may otherwise be stifled by the conservative nature of established systems.

Accelerated feature development: Forks provide a pathway for implementing features that may be controversial or risky. Instead of requiring the entire community to adopt changes that some members view with scepticism, forking allows advocates of new features to pursue their vision independently. If successful, these innovations may eventually be adopted by the original chain or inspire further developments across the ecosystem.

Specialised use cases: As blockchain technology matures, different sectors and applications require specialised features. Forking allows blockchain protocols to be tailored to specific use cases without compromising the original chain's focus. For example, a blockchain optimised for financial transactions may fork to create a version better suited for supply chain management or digital identity verification.

Competition as innovation driver: The competitive dynamics created by forking can spur innovation on both the original chain and its forks. When a prominent fork challenges an established blockchain, it often prompts the original project to accelerate its own development roadmap and address community concerns more proactively.

Community empowerment: Perhaps most significantly, forking empowers blockchain communities by providing an exit option when governance mechanisms fail to resolve differences. This 'vote with your code' approach ensures

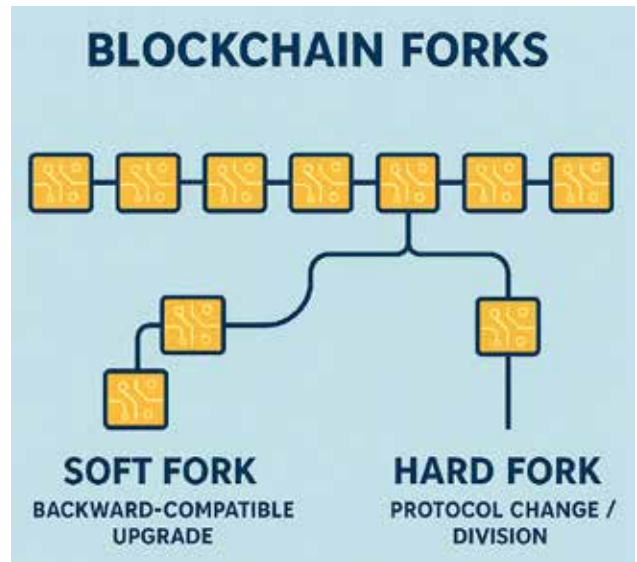


Figure 2: Forks

that no single entity can indefinitely control a blockchain's development direction, reinforcing the decentralised ethos that underlies the technology.

Successful forks have introduced groundbreaking innovations, from new consensus mechanisms to scalability solutions that have advanced the entire field. Even when forks ultimately fail to gain widespread adoption, the experimentation they facilitate generates valuable knowledge that benefits the broader ecosystem.

The risks of fragmentation

While forking enables innovation, it also introduces significant risks of ecosystem fragmentation that can undermine the long-term viability and adoption of blockchain technology.

Diluted network effects: Blockchain networks derive much of their value from network effects—the principle that a network becomes more valuable as more users join it. When a blockchain forks, it divides its user base, potentially reducing the utility of both resulting networks. This fragmentation can weaken critical metrics such as:

- Security (through reduced hash power or validator participation)
- Liquidity in DeFi applications
- Developer resources and community support
- Market recognition and trust

User confusion and adoption barriers: For mainstream users and institutions considering blockchain adoption, the proliferation of forks creates significant confusion. Which version of a blockchain should they trust? How can they evaluate the security and longevity of competing chains? This uncertainty can delay adoption decisions and drive potential